Chapter 4
Advection–Diffusion Problems

In this chapter, the physics of advection and advection diffusion will be explained. Lattice Boltzmann method will be discussed for solving different advection–diffusion problems for one and two dimensional cases. Extending the method to three dimension problems is straightforward.

4.1 Advection

The advection is the transport of heat, mass, and momentum by bulk motion of fluid particles. The governing advection equation for one dimensional (1-D) in Cartesian coordinate system can be expressed as,

$$\frac{\partial \phi}{\partial t} + u\frac{\partial \phi}{\partial x} = 0. \tag{4.1}$$

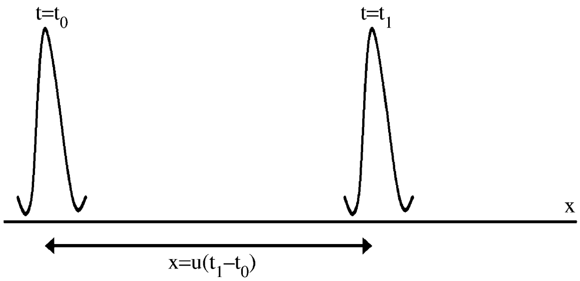
where ϕ is a dependent variable (mass, momentum, energy, species, etc.) and u is the advection velocity, fluid flow velocity. The above equation has first order derivative in space and in time; hence it needs only a boundary condition and an initial condition for closing the solution. Equation 4.1 conveys us that the dependent variable, ϕ, advects with u-velocity in space as the time proceeds. The property ϕ is not diffused or distorted, but it just translates with speed u. For instance, if the property ϕ has a given profile as shown in Fig. 4.1 at time, $t = t_0$, then after $t = t_1$, the profile conserves its original shape and only translates (advectes) with a given velocity u, to a different location.

If u is not a function of ϕ, then the above equation is linear and analytical solution can be obtained. For instance if $\phi(0, x) = g(x)$, then the solution is $g(x - ut)$, where $g(x)$ is the profile of the function at $t = 0$.

As an example, if a drop of oil is placed on a water stream, the drop of oil advects without any diffusion. It conserves its shape and density as it moves with the velocity of the water stream.

DOI: 10.1007/978-0-85729-455-5_4,

Fig. 4.1 Illustrates advection processes



Three dimensional advection equation can be written as

$$\frac{\partial \phi}{\partial t} + u\frac{\partial \phi}{\partial x} + v\frac{\partial \phi}{\partial y} + w\frac{\partial \phi}{\partial z} = 0 \tag{4.2}$$

which is an extension for 1-D equation.

4.2 Advection–Diffusion Equation

The advection–diffusion process is a process where both advection and diffusion take place simultaneously. For instance, if a pollutant or a drop of ink is added to a stream of water, the pollutant or ink concentration decreases (diffuses) as the stream moves away from the source. The phenomenon of advection–diffusion without and with a source or reaction term is very common in the nature and in industrial and engineering applications, usually called transport problem.

The one-dimensional advection–diffusion equation can be expressed in Cartesian coordinate system as,

$$\frac{\partial \phi}{\partial t} + u\frac{\partial \phi}{\partial x} = \alpha\frac{\partial^2 \phi}{\partial x^2} \tag{4.3}$$

where α is the diffusion coefficient as discussed in the previous chapter.

Figure 4.2 illustrates the concept of advection–diffusion process. Initial function propagates in space as the time proceeds, at the same time the shape of the function changes (flatten) due to diffusion process.

4.2.1 Finite Difference Method

In numerical methods, advection term may cause problem if not properly approximated. The flow direction affect the information flow, upstream should have effect on the downstream, while effect of the downstream condition on the

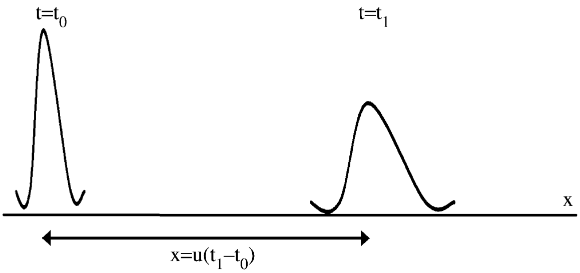


Fig. 4.2 Advection–diffusion processes

upstream condition depends on the strength of the flow (magnitude of flow velocity). In more precious wording, it depends on ratio of advection to diffusion, Peclet number. Hence, using central difference,

$$\left(\frac{\phi(i+1)-\phi(i-1)}{2\Delta x}\right),$$

to approximate the first derivative in space $(\frac{\partial\phi}{\partial x})$ may lead to unstable solution, because we are giving the same weight for both, left and right hand of the function ϕ. In other words, we are not taking into consideration that more information is transferring in the stream-wise direction. In order to take into the consideration the information flow direction, it is more appropriate to use up-wind scheme,

$$\frac{\partial\phi}{\partial x}=\begin{cases}\frac{\phi(i)-\phi(i-1)}{\Delta x} & \text{if } u>0\\ \frac{\phi(i+1)-\phi(i)}{\Delta x} & \text{if } u\le 0\end{cases} \tag{4.4}$$

Note that Eq. 4.4 is a first order accurate and in general causes artificial diffusion (false diffusion). In other words, the accuracy of the solution depends on the ratio of a numerically introduced diffusion coefficient to the physical diffusion coefficient (α). The remedy for this problem is to use higher order up-wind schemes, which need more nodes to be involved in Eq. 4.4. For more detail, we refer to textbooks on numerical methods for computational fluid dynamics (CFD).

The diffusion term (second order term) can be approximated, as discussed in Chap. 3, i.e., using the central difference scheme.

The code for explicit, finite difference for positive $u = 0.1$ is given in the appendix. Initially, the value of ϕ is zero through the domain and instantly the left-hand side boundary condition for ϕ is set to unity.

For stability reason, Δt and Δx are set to 0.25 and 0.5, respectively. Since u is positive, the first derivative is approximated as it is given in the first line of Eq. 4.4.

Forward difference is used in approximating the time derivative. Again the approximation is first order accurate. Better approximation is possible, such as using Crank–Nikelson method. For details, we refer to numerical method textbooks on the subject.

4.2.2 The Lattice Boltzmann

4.2.2.1 The Lattice Boltzmann Method for Advection–Diffusion Problems

The Lattice Boltzmann equation for advection–diffusion problem is the same as Eq. 3.18,

$$f_k(x+\Delta x, t+\Delta t) = f_k(x,t)[1-\omega] + \omega f_k^{\mathrm{eq}}(x,t) \tag{4.5}$$

The only difference is an extra term added to the equilibrium distribution function to accommodate the advection effect,

$$f_k^{\mathrm{eq}} = w_k \phi(x,t)\left[1 + \frac{c_k \cdot \vec{u}}{c_s^2}\right] \tag{4.6}$$

where $\vec{u}$ is advection velocity vector. For example for 2-D, the vector $\vec{u} = ui + vj$, where i and j are unit vectors along x and y-direction. And c_k is unit vector along the streaming direction. In general,

$$c_k = \frac{\Delta x}{\Delta t} i + \frac{\Delta y}{\Delta t} j \tag{4.7}$$

For $c_k = 1$, the speed of sound, c_s, is as follows: for D1Q2, D2Q4, and D3Q6,

$$c_s = \frac{1}{\sqrt{2}} \tag{4.8}$$

for D1Q3, D2Q5, D2Q9, and D3Q15,

$$c_s = \frac{1}{\sqrt{3}} \tag{4.9}$$

Same procedure can be followed for diffusion problem. The LBM code is the same as for diffusion problem (Chap. 3), except that an extra term is need to be added to f^{eq}.

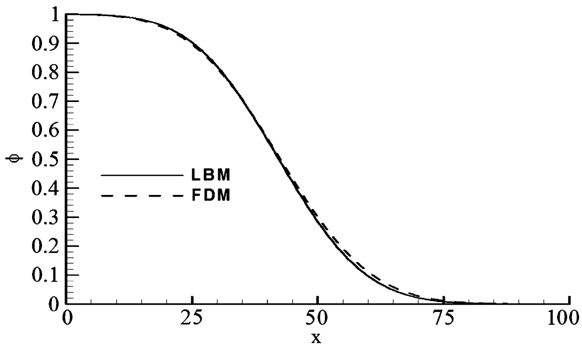


Fig. 4.3 Comparison between predictions of LBM and FDM for advection–diffusion problem

Computer code for 1-D advection–diffusion equation can be found in the appendix for the same problem as we discussed for finite different method.

Figure 4.3 compares the results obtained by using finite difference and LBM method for the mentioned problem. Note that Δx used for FDM is 0.5 compared with $\Delta x = 1.0$ used for LBM to get same results. LBM method is more stable and produce accurate results compared with FDM. FDM needs smaller Δx to produce accurate results, due to numerical errors mentioned before.

4.3 Equilibrium Distribution Function

For advection diffusion problem, it is appropriate to assume that the equilibrium distribution functions can be approximated as a polynomial in velocity. Let,

$$f_i^{\text{eq}} = A_i + B_i c_i \cdot u \tag{4.10}$$

Equilibrium distribution function must satisfy the conservation of mass and momentum,

$$\sum_{i=1}^{2} f_i^{\text{eq}} = \Theta \tag{4.11}$$

and

$$\sum_{i=1}^{2} f_i^{\text{eq}} c_i = \Theta u \tag{4.12}$$

Hence Eq. 4.11 yields,

$$\sum_{i=1}^{2} A_i = A_1 + B_1 u + A_2 - B_2 u = \Theta \tag{4.13}$$

This implies that,

$$A_1 + A_2 = \Theta \tag{4.14}$$

and

$$B_1 u - B_2 u = 0 \quad \text{or} \quad B_1 = B_2 \tag{4.15}$$

Eq. 4.12,

$$\sum_{i=1}^{2} (A_i c_i + B_i c_i c_i u) = u\Theta \tag{4.16}$$

where $c_1 = 1$ and $c_2 = -1$. By comparing coefficients of both sides of Eq. 4.19, we obtain,

$$\sum_{i=1}^{2} A_i c_i = 0 \tag{4.17}$$

Hence, $A_1 = A_2$. In general,

$$A_i = \omega_i \Theta \tag{4.18}$$

where $\omega_i = A_i/\Theta$. Also, from Eq. 4.19,

$$B_i c_i c_i = \Theta \tag{4.19}$$

Multiply both sides of Eq. 4.19 by ω_i and from Chapman–Enskog expansion, the following relation can be obtained,

$$w_i c_i c_i = \frac{1}{c_s^2} \tag{4.20}$$

we can get the following relation,

$$B_i = \frac{w_k}{c_s^2} \Theta \tag{4.21}$$

Hence,

$$f_i^{eq} = w_i \Theta \left(1 + \frac{u c_i}{c_s^2}\right) \tag{4.22}$$

In order to relate LBM to macro scale problem, multi-scale expansion is needed. In the following sections, Chapman–Enskog expansion for advection–diffusion problem will be discussed in detail.

4.4 Chapman–Enskog Expansion

For example, the advection–diffusion equation for energy equation without source term (temperature) in Cartesian coordinate can be written as,

$$\frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} = \Gamma \frac{\partial^2 T}{\partial x^2} \tag{4.23}$$

where $T(x, t)$ is dependant parameter (T stands for temperature for energy equation, for velocity for momentum equation, or for species concentration for species conservation equation). It is assumed that diffusion coefficient, Γ, is a constant. The advection–diffusion equation can be scaled spatially as, x is set to x/ϵ, where ϵ is a small parameter. Introducing the scale to advection diffusion equation yields,

$$\frac{\partial T(x,t)}{\partial t} + u\epsilon \frac{\partial T(x,t)}{\partial x} = \Gamma \epsilon^2 \frac{\partial^2 T(x,t)}{\partial x^2} \tag{4.24}$$

Now, if we scale time t, with $1/\epsilon^2$ as we did in Chap. 3 for diffusion equation, then the first term on the left-hand side of the equation balances the diffusion term on the right-hand side of the equation, hence the effect of advection term will be neglected. If we scale time t, with $1/\epsilon$, in this case there will be a balance between the first and second terms on the left-hand side of the equation, and the effect of diffusion term will be neglected. To resolve this issue,we need to introduce dual time scales, i.e.,

$$\frac{\partial}{\partial t} = \epsilon \frac{\partial}{\partial t_1} + \epsilon^2 \frac{\partial}{\partial t_2} \tag{4.25}$$

Hence, the first term in the above equation balances the advection term and the second term balances the diffusion term. The problem is resolved.

For one-dimensional problem with two lattice velocity components $(c_i, i = 1,2)$, the temperature can be expressed as,

$$T(x,t) = \sum_{i=1}^{i=2} f_i(x,t) = f_1(x,t) + f_2(x,t) \tag{4.26}$$

From now and on, we write f_i instead of $f_i(x,t)$ for simplicity. The equilibrium distribution function can be expressed as,

$$f_i^{\mathrm{eq}} = w_i T(x,y)(1 + Ac_i \cdot u) \tag{4.27}$$

where A is a constant. The second term on the right-hand side accounts for advection effect. By taking summation of both sides of the above equation,

$$\sum_i f_i^{\mathrm{eq}} = T = w_1 T + w_2 T \tag{4.28}$$

Notice that the term $Ac_i \cdot u$ cancels in the summation process because $c_1 = 1$ and $c_2 = -1$.

The weight factors must satisfy the relation

$$\sum_{i=1}^{i=2} w_i = 1 \tag{4.29}$$

i.e., $w_1 + w_2 = 1$, which yields, $w_1 = w_2 = 1/2$.

The distribution function can be expanded in terms of a small parameter, ϵ as

$$f_i(x,t) = f_i^0 + \epsilon f_i^1 + \epsilon^2 f_i^2 + \cdots \tag{4.30}$$

where f_i^0 is distribution function at the equilibrium conditions, equal to f_i^{eq} By summing above equation, it leads to,

$$\sum_{i=1}^{i=2} f_i(x,t) = \sum_{i=1}^{i=2} [f_i^0 + \epsilon f_i^1 + \epsilon^2 f_i^2 + \cdots] \tag{4.31}$$

Since, $f_1 + f_2 = T(x,t)$ and $f_1^0 + f_2^0 = T(x,t)$, hence other expanded term in the above equation should be zero, i.e,

$$\sum_{i=1}^{i=2} f_i^1 = 0 \tag{4.32}$$

$$\sum_{i=1}^{i=2} f_i^2 = 0 \tag{4.33}$$

The updated distribution function $f_i(x + c_i\Delta t, t + \Delta t)$ is expanded using Taylor series,

$$\begin{aligned} f_i(x + c_i\Delta t, t + \Delta t) = & f_i(x,t) + \frac{\partial f_i}{\partial t}\Delta t + \frac{\partial f_i}{\partial x} c_i \Delta t \\ & + 1/2\Delta t^2 \left(\frac{\partial^2 f_i}{\partial t^2} + 2\frac{\partial^2 f_i}{\partial t \partial x} c_i + \frac{\partial^2 f_i}{\partial x^2} c_i c_i \right) + O(\Delta t)^3 \end{aligned} \tag{4.34}$$

Introducing scaling for the above equation, i.e., $\frac{\partial}{\partial t}$ is replaced by $\epsilon \frac{\partial}{\partial t_1} + \epsilon^2 \frac{\partial}{\partial t_2}$ and $\frac{1}{\partial x}$ is replaced by $\epsilon \frac{1}{\partial x}$, by keeping terms with ϵ^2, yields:

$$\begin{aligned} f_i(x + c_i\Delta t, t + \Delta t) = & f_i(x,t) + \epsilon \frac{\partial f_i}{\partial t_1}\Delta t + \epsilon^2 \frac{\partial f_i}{\partial t_2}\Delta t + \epsilon \frac{\partial f_i}{\partial x} c_i \Delta t \\ & + \frac{\Delta t^2}{2} \left(\frac{\epsilon^2 \partial^2 f_i}{\partial t_1^2} + 2\frac{\epsilon^2 \partial^2 f_i}{\partial t_1 \partial x} c_i + \frac{\epsilon^2 \partial^2 f_i}{\partial x^2} c_i c_i \right) + O(\Delta t)^3 \end{aligned} \tag{4.35}$$

Lattice Boltzmann equation can be written as,

$$f_i(x + c_i\Delta t, t + \Delta t) = f_i(x,t) + \frac{\Delta t}{\tau}[f_i^0(x,t) - f_i(x,t)] \tag{4.36}$$

Substituting Eq. 4.35 into above equation and retaining terms up to order of ϵ^2, yields:

$$\begin{aligned} \frac{1}{\tau}[f_i^0(x,t) - f_i(x,t)] = & \epsilon \frac{\partial f_i}{\partial t_1} + \epsilon^2 \frac{\partial f_i}{\partial t_2} + \epsilon \frac{\partial f_i}{\partial x} c_i + \frac{\epsilon^2 \Delta t}{2} \left(\frac{\partial^2 f_i}{\partial t_1^2} + 2\frac{\partial^2 f_i}{\partial x \partial t_1} + \frac{\partial^2 f_i}{\partial x^2} c_i c_i \right) \\ & + O(\Delta t)^2 + O(\epsilon^3) \end{aligned} \tag{4.37}$$

where f_i^0 is f_i^{eq}. Expanding $f_i = f_i^0 + \epsilon f_i^1 + \epsilon^2 f_i^2 + \cdots$ and substituting in the above equation, yields,

$$
\begin{aligned}
-\frac{1}{\tau}\left(\epsilon f_i^1+\epsilon^2 f_i^2+\cdots\right) &= \epsilon\frac{\partial f_i^0}{\partial t_1}+\epsilon^2\frac{\partial f_i^1}{\partial t_1}+\epsilon^2\frac{\partial f_i^0}{\partial t_2}+\epsilon\frac{\partial f_i^0}{\partial x}c_i+\epsilon^2\frac{\partial f_i^1}{\partial x}c_i \\
&+\frac{\Delta t\epsilon^2}{2}\left[\frac{\partial^2 f_i^0}{\partial t_1^2}+2\frac{\partial^2 f_i^0}{\partial x\partial t_1}c_i+\frac{\partial f_i^0}{\partial x}c_ic_i\right]+O(\epsilon^3)+)(\Delta t^2)
\end{aligned}
\tag{4.38}
$$

Collecting the terms of order of ϵ from both sides of the above equations:

$$-\frac{1}{\tau}f_i^1=\frac{\partial f_i^0}{\partial t_1}+\frac{\partial f_i^0}{\partial x}c_i \tag{4.39}$$

Latter on we may need derivative of f_i^1 with respect to t_1 and x, hence by taking the derivative of above equation, respectively:

$$-\frac{1}{\tau}\frac{\partial f_i^1}{\partial t_1}=\frac{\partial^2 f_i^0}{\partial t_1^2}+\frac{\partial^2 f_i^0}{\partial t_1\partial x}c_i \tag{4.40}$$

and

$$-\frac{1}{\tau}\frac{\partial f_i^1}{\partial x}=\frac{\partial^2 f_i^0}{\partial t_1\partial x}+\frac{\partial^2 f_i^0}{\partial x^2}c_i \tag{4.41}$$

Multiply Eq. 4.41 by c_i, add it to Eq. 4.40 and rearrange the summated equation,

$$-\frac{1}{\tau}\left(\frac{\partial f_i^1}{\partial t_1}+\frac{\partial f_i^1}{\partial x}c_i\right)=\frac{\partial^2 f_i^0}{\partial t_1^2}+2\frac{\partial^2 f_i^0}{\partial t_1\partial x}c_i+\frac{\partial^2 f_i^0}{\partial x^2}c_ic_i \tag{4.42}$$

By integrating Eq. 4.40 and taking the summing over all values of i ($i=1, 2$) we obtain:

$$-\frac{1}{\tau}\sum_{i=1}^{2}f_i^1=\sum_{i=1}^{2}\frac{\partial f_i^0}{\partial t_1}+\sum_{i=1}^{2}\frac{\partial f_i^0}{\partial x}c_i \tag{4.43}$$

The term on the left-hand side diminishes (equal to zero). The summation of f_i^0 is equal to T. Also,$\sum_{i=1}^{2}f_i^0c_i=uT$. Hence the above equation, reformulated as,

$$0=\frac{\partial T}{\partial t_1}+\frac{\partial(uT)}{\partial x} \tag{4.44}$$

which is advection equation. Notice that t_1 balance advection term.

Now let us return to Eq. 4.38 and collect order terms of ϵ^2 :

$$-\frac{1}{\tau}f_i^2=\frac{\partial f_i^1}{\partial t_1}+\frac{\partial f_i^0}{\partial t_2}+\frac{\partial f_i^1}{\partial x}c_i+\frac{\Delta t}{2}\left[\frac{\partial^2 f_i^0}{\partial t_1^2}+2\frac{\partial^2 f_i^0}{\partial x\partial t_1}c_i+\frac{\partial^2 f_i^0}{\partial x^2}c_ic_i\right] \tag{4.45}$$

Substituting Eq. 4.38 into the above equation, yields:

$$-\frac{1}{\tau}f_i^2 = \frac{\partial f_i^1}{\partial t_1} + \frac{\partial f_i^0}{\partial t_2} + \frac{\partial f_i^1}{\partial x}c_i - \frac{\Delta t}{2\tau}\left[\frac{\partial f_i^1}{\partial t_1} + \frac{\partial f_i^1}{\partial x}c_i\right] \tag{4.46}$$

Collecting similar terms,

$$-\frac{1}{\tau}f_i^2 = \frac{\partial f_i^0}{\partial t_2} + (1 - \frac{\Delta t}{2\tau})\left[\frac{\partial f_i^1}{\partial t_1} + \frac{\partial f_i^1}{\partial x}c_i\right] \tag{4.47}$$

Summing the above equation over all states, $i = 1, 2$,

$$\sum_i \left[-\frac{1}{\tau}f_i^2\right] = \sum_{i=1}^{2}\left[\frac{\partial f_i^0}{\partial t_2} + \left(1 - \frac{\Delta t}{2\tau}\right)\right]\left[\sum_{i=1}^{2}\frac{\partial f_i^1}{\partial t_1} + \sum_{i=1}^{2}\frac{\partial f_i^1}{\partial x}c_i\right] \tag{4.48}$$

The left-hand side (LHS) is zero. The first term on the right-hand side (RHS), will be

$$\sum_{i=1}^{2}\frac{\partial f_i^0}{\partial t_2} = \frac{\partial \sum_{i=1}^{2} f_i^0}{\partial t_2} = \frac{\partial \Theta}{\partial t_2} \tag{4.49}$$

The term of,

$$\sum_i \left[\frac{\partial^2 f_i^0}{\partial x^2}c_i c_i\right] = \frac{\partial^2}{\partial x^2}\left(\sum_i [f_i^0 c_i c_i]\right) = \frac{\partial^2 \Theta}{\partial x^2} \tag{4.50}$$

Hence Eq. 4.48 can be simplified as,

$$0 = \frac{\partial \Theta}{\partial t} - \left(\tau - \frac{\Delta t}{2}\right)\frac{\partial^2 \Theta}{\partial x^2} \tag{4.51}$$

By comparing above equation with continuum diffusion equation, the relaxation parameter, τ, can be related to the diffusion coefficient as,

$$\Gamma = \tau - \frac{\Delta t}{2} \tag{4.52}$$

4.4.1 Two-Dimensional Advection–Diffusion Problems

$$\frac{\partial \phi}{\partial t} + u\frac{\partial \phi}{\partial x} + v\frac{\partial \phi}{\partial y} = \alpha\left(\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2}\right) \tag{4.53}$$

The finite difference approximation of the above equation is straightforward for a given velocity field. However, care should be taken if the velocity direction is changing.

First order accurate up-wind scheme can be written as,

$$\frac{\partial\phi}{\partial x} = \begin{cases} \frac{\phi(i,j)-\phi(i-1,j)}{\Delta x} & \text{if } u > 0 \\ \frac{\phi(i+1,j)-\phi(i,j)}{\Delta x} & \text{if } u \le 0 \end{cases} \tag{4.54}$$

and

$$\frac{\partial\phi}{\partial y} = \begin{cases} \frac{\phi(i,j)-\phi(i,j-1)}{\Delta y} & \text{if } v > 0 \\ \frac{\phi(i,j+1)-\phi(i,j)}{\Delta y} & \text{if } v \le 0 \end{cases} \tag{4.55}$$

As an example, let us consider flow in a square porous domain shown in Fig. 4.4. Initially the domain is at zero temperature. The upper boundary of the domain is kept at zero temperature, while the bottom and vertical right boundaries are assumed to be adiabatic. The left vertical wall is subjected to unity temperature at time $= 0$. Find the temperature distribution in the domain at time $= 200$, for material of diffusivity of 1.0. The velocity field of $u = 0.1$ and $v = 0.2$ is kept constant in the domain.

For finite difference it is sufficient to use rectangular grids of $\Delta x = 1.0$ and $\Delta y = 1.0$.

To ensure stable solution $\Delta t = 0.2$, for FDM. The computer code is given in Appendix.

4.5 Two-Dimensional Lattice Boltzmann Method

In the following sections, two approaches will be discussed, namely D2Q4 and D2Q9.

4.5.1 D2Q4

The lattice arrangement for D2Q4 is shown in Fig. 4.5. The method is similar to LBM introduced to solve diffusion equation except that the equilibrium term, as mentioned before.

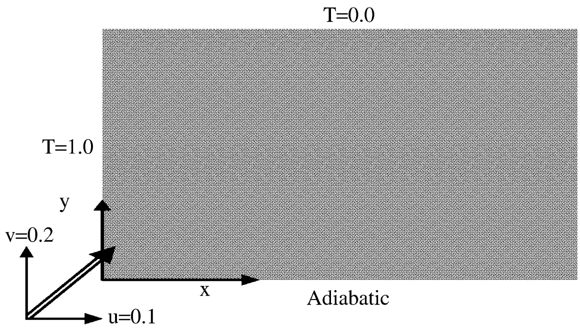


Fig. 4.4 Flow and heat transfer in a porous medium

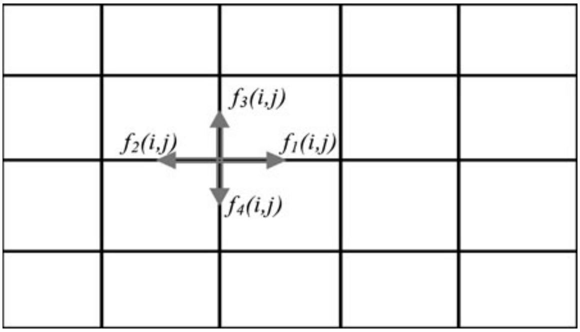


Fig. 4.5 Lattice arrangement for D2Q4

The Lattice BGK equation,

$$f_k(x + \Delta x, t + \Delta t) = f_k(x, t)[1 - \omega] + \omega f_k{}^{\mathrm{eq}}(x, t) \tag{4.56}$$

The k takes from 1 to 4.

The streaming and collision processes are similar to what is explained in previous chapter, Chap. 3.

The equilibrium part can be calculated using Eq.4.6,

$$f_k^{\mathrm{eq}} = w_k \phi(x, t)\left[1 + \frac{c_k \cdot \vec{u}}{c_s^2}\right] \tag{4.57}$$

where $w_k = 0.25$ for $k = 1, 4$

For $\Delta x = \Delta y = \Delta t = 1.0$

$$\frac{c_k \cdot \vec{u}}{c_s} = \begin{cases} 2u & k = 1 \\ -2u & k = 2 \\ 2v & k = 3 \\ -2v & k = 4 \end{cases} \tag{4.58}$$

Notice that c_k is unit vector along the streaming line and

$$c_s = \frac{c_k}{\sqrt{2}} \tag{4.59}$$

for D2Q4.

The computer code (see Appendix) is clear and there is no need to elaborate more on the streaming and collision processes evaluations because it is similar to procedure discussed in Chap. 3. Also, setting boundary conditions is same as in Sect. 3.6.4. Figure 4.6 shows the temperature profile at the mid-plane $y = 0.5$ predicted by LBM and FDM.

4.5.2 D2Q9

The lattice arrangement for D2Q9 is shown below Fig. 4.7 which is the same as in Fig. 3.12.

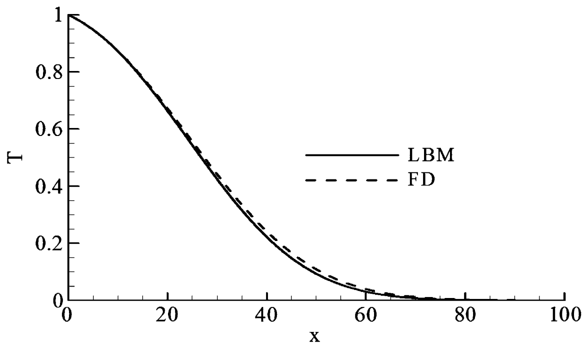


Fig. 4.6 Temperature profile at the mid-plane ($y = 0.5$) predicted by LBM and FDM

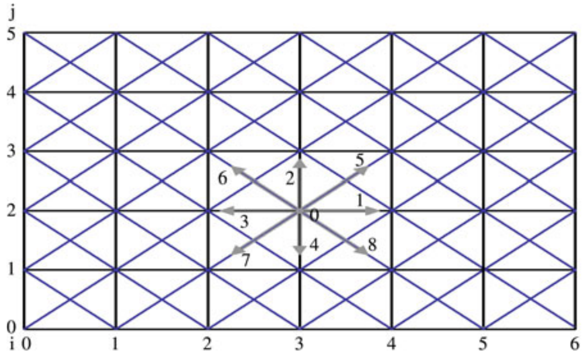


Fig. 4.7 Lattice arrangement for D2Q9

The procedure is the same as discussed in Sect. 3.4.3, except the f^{eq} should account for the advection term,

$$f_k^{\mathrm{eq}} = w_k \phi(x,t)\left[1 + \frac{c_k \cdot \vec{u}}{c_s^2}\right] \tag{4.60}$$

Hence,

$$\begin{aligned}
f_1^{\mathrm{eq}} &= w_1 \phi(1.0 + 3.0u) \\
f_2^{\mathrm{eq}} &= w_2 \phi(1.0 + 3.0v) \\
f_3^{\mathrm{eq}} &= w_3 \phi(1.0 - 3.0u) \\
f_4^{\mathrm{eq}} &= w_4 \phi(1.0 - 3.0v) \\
f_5^{\mathrm{eq}} &= w_5 \phi(1.0 + 3.0(u + v))
\end{aligned}$$

etc.

For details of implementation and coding, see the code for D2Q9 in the appendix. Notice that in programming, $f_1, f_2, f_3, \ldots$ used with subscripts for two reasons, compact programming and the code can be easily extended to 3-D. The code simulates heat transfer as in previous problem, with $v = 0.4$ and $u = 0.1$. The conditions on all boundaries are kept at zero temperature, except the left side is suddenly changed to unity. The thermal diffusivity of the medium is set to 1.0. The simulation is carried for 400 time steps. The code is given in Appendix.

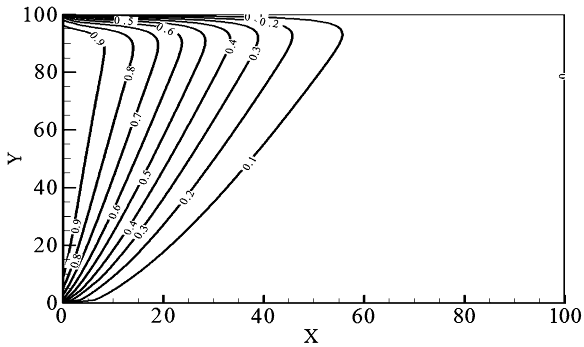


Fig. 4.8 Isotherms at time = 400

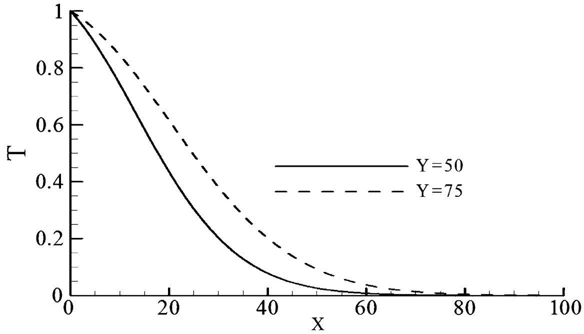


Fig. 4.9 Temperature profiles at the mid-height ($Y = 50$) and at the third quarter height ($Y = 75$) for time = 400

Figure 4.8 shows the isotherm plot at time = 400. Figure 4.9 shows the temperature profiles at mid-height ($Y = 50$) and at height $Y = 75$, for time = 400.

Note that, the scale of the dimension of the problem can be non-dimensionalized or scaled after the calculation. For instance, the height and length of the domain can be divided to the total height of the domain, velocity field can be divided to the inlet velocity, and so on.

At this stage the reader should feel comfortable to solve advection–diffusion problems with and without source terms. The source or sink term can be added as discussed in Chap. 3.

4.6 Problems

4.6.1 Combustion in Porous Layer

Porous burner, or combustion in porous media can be model as a two separate energy equations, one for solid phase and other for gaseous phase. Assume a premixed gaseous fuel with air flows through a porous layer, as shown in Fig. 4.10 combustion takes place inside the layer. For simplicity, let us model the combustion as a constant heat source term released at certain location in the porous medium.

The following model equations need to be solved;

Energy equation for gaseous phase:

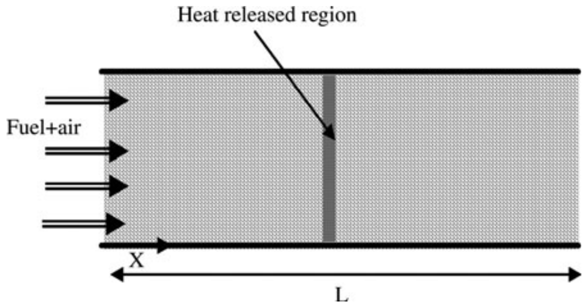


Fig. 4.10 Schematic diagram of porous burner

$$\frac{\partial(\rho_g c_g T_g)}{\partial t} + u\frac{\partial(\rho_g c_g T_g)}{\partial x} = \frac{\partial}{\partial x}\left(k_g \frac{\partial T_g}{\partial x}\right) - h_v(T_g - T_s) + S \tag{4.61}$$

Energy equation for solid phase:

$$\frac{\partial(\rho_s c_s T_s)}{\partial t} = \frac{\partial}{\partial x}\left(k_s \frac{\partial T_s}{\partial x}\right) + h_v(T_g - T_s) \tag{4.62}$$

where ρ, c, T, and h_v are density, specific heat, temperature, and volumetric heat transfer coefficient, respectively. The subscript g and s stand for gas and solid phases, respectively. The source term, S, has value only at the location of heat release and zero at other locations.

The boundary condition:

At inlet, $x = 0$, for simplicity assume that $T_g = T_s = T_{in}$ and at the outlet, $x = L$, assume that the gradient of the temperatures are zero. Modify above code to simulate a flow in a porous layer.

Hint use two distribution functions, one for gas temperature and other for solid temperature.

4.6.2 Cooling a Heated Plate

1. Fully developed flow between parallel plates can be assumed as $u/u_{in} = 1.5(1 - y/H)$. A cold flow with temperature of 20°C enters the channel. The bottom of the channel maintained at $T = 80$°C, while the top plate is assumed to be perfectly insulated. Calculate the temperature field between the plates and rate of heat transfer from the bottom plate (Fig. 4.11).

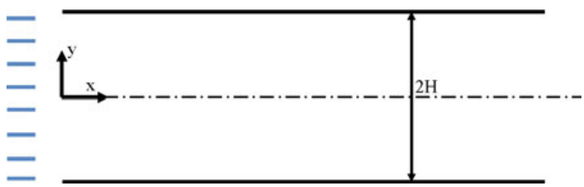


Fig. 4.11 Forced convection problem diagram

4.6.3 Coupled Equations with Source Term

In this section we consider advection–diffusion equation with a source term, which arises in modeling Soret effect. The non-dimensional energy and species conservation equations can be written as,

$$\frac{\partial \Theta}{\partial t} + u\frac{\partial \Theta}{\partial x} = \frac{1}{Pr}\frac{\partial^2 \Theta}{\partial x^2} \tag{4.63}$$

and

$$\frac{\partial \Phi}{\partial t} + u\frac{\partial \Phi}{\partial x} = \frac{1}{PrSc}\left[\frac{\partial^2 \Phi}{\partial x^2} - \frac{\partial^2 \Theta}{\partial x^2}\right] \tag{4.64}$$

respectively. The source term is the second term on the right-hand side of Eq. 4.64 ($\frac{1}{PrSc}\frac{\partial^2 \Theta}{\partial x^2}$) is the source (or sink) term need to be modelled correctly. Hence to solve this kind of problem, it is necessary to use two distribution functions, one for energy equation and another for species conservation equation, such as f_i and g_i for temperature and species concentration, respectively. The source term can be added to the species equation as,

$$S_i = w_i \, \Delta t \, S \tag{4.65}$$

where $S = -\frac{1}{PrSc}\frac{\partial^2 \Theta}{\partial x^2}$ and w_i is the weighting coefficient. The second derivative can be approximated by using second order central difference. Use D1Q2 and D1Q3 to solve the above problem for $Pr = 1.2$ and $Sc = 2.5$ for a given velocity $u = 0.5$. The flow initially was at zero temperature and zero species concentration. Then, suddenly a hot and polluted fluid with $\Theta = 1.0$ and $\Phi = 1.0$ forced to the domain. Compare results of D1Q2 and D1Q3. You will notice that prediction of D1Q2 may be oscillatory, which is physically incorrect. We found that D1Q2 scheme may produce oscillatory solution with source term, which is not the case for D1Q3. Therefore, it is recommended to use D1Q3 for 1-D and D2Q5 for 2-D problems.